Marmara Üniversitesi
Electrical and Electronics Engineering Department
Fall 2023
EE 3016/3051 – Quiz 1
November 7, 2023

Name: Key	ID							5 (Vector)	3 (Point)

$$\vec{D} = \vec{a}_x - (2x+y)\,\vec{a}_y + y\,\vec{a}_z$$
$$P(1,0,-2)$$

(1) $\vec{D} = \vec{a}_x - (2x+y)\,\vec{a}_y + y\,\vec{a}_z$

$\Psi_{top} + \Psi_{bottom} = 0$ & $\Psi_{back} + \Psi_{front} = 0$

$$\Psi_{left} = \iint_{left} \vec{D}\cdot(-dx\,dz\,\vec{a}_y) = \int_{-1}^{1}\int_{1}^{3}\underset{(y=0)}{(2x+y)}\,dx\,dz$$

$$= 2\,\frac{x^2}{2}\Big|_1^3\, z\Big|_{-1}^{1} = 16 \quad (C)$$

$$\Psi_{right} = \iint \vec{D}\cdot(dx\,dz\,\vec{a}_y) = \int_{-1}^{1}\int_{1}^{3}\underset{(y=1)}{-(2x+y)}\,dx\,dz$$

$$= -2\,\frac{x^2}{2}\Big|_1^3\, z\Big|_{-1}^{1} - x\Big|_1^3\, z\Big|_{-1}^{1} = -16 - 4 \quad (C)$$

$\Psi_{total} = -4$ (C)

Another method

Use divergence theorem

$$flux = \int_V (\vec{\nabla}\cdot\vec{D})\,dv$$

$$\vec{\nabla}\cdot\vec{D} = \frac{\partial}{\partial x}(1) - \frac{\partial}{\partial y}(2x+y) + \frac{\partial}{\partial z}(y) = -1$$

$$flux = \int_{-1}^{1}\int_{0}^{1}\int_{1}^{3}(-1)\,dx\,dy\,dz = -4 \quad (C)$$

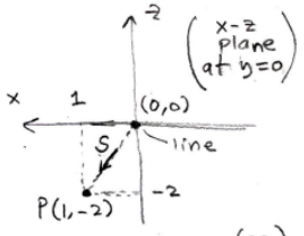
(2) $P(1,0,-2)$

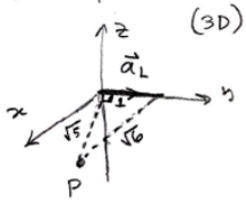
$s = \sqrt{1+2^2} = \sqrt{5}$; $\vec{a}_L = \vec{a}_y$

$\vec{a}_n = (\vec{a}_x + 2\vec{a}_z)/\sqrt{5}$

$\alpha_1 = 0 \Rightarrow \sin\alpha_1 = 0$; $\cos\alpha_1 = 1$

$\sin\alpha_2 = 1/\sqrt{6}$; $\cos\alpha_2 = \sqrt{5}/\sqrt{6}$





$$E_P = \frac{-1\times10^{-3}}{4\pi\epsilon\sqrt{5}}\left[\frac{1}{\sqrt{6}}\,\frac{\vec{a}_x - 2\vec{a}_z}{\sqrt{5}} - \left(\frac{\sqrt{5}}{\sqrt{6}} - 1\right)\vec{a}_y\right] \quad (V/m)$$